Mathematics Formula Handbook

Every formula includes why it exists and when to reach for it

1. Standard Derivatives

Use when: You need the derivative of a single named function (power, exponential, trig, log, inverse trig, or hyperbolic) as a building block inside a bigger differentiation problem. Memorize these – almost every calculus question uses several of them in combination.

$$\frac{d}{dx}(x^n) = nx^{n-1}, \quad \frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(a^x) = a^x \log a$$
$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$
$$\frac{d}{dx}(\tan x) = \sec^2 x, \quad \frac{d}{dx}(\cot x) = -\csc^2 x$$
$$\frac{d}{dx}(\sec x) = \sec x \tan x, \quad \frac{d}{dx}(\csc x) = -\csc x \cot x$$
$$\frac{d}{dx}(\log x) = \frac{1}{x}, \quad \frac{d}{dx}(\log_a x) = \frac{\log_a e}{x}$$
$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$
$$\frac{d}{dx}(\sinh x) = \cosh x, \quad \frac{d}{dx}(\cosh x) = \sinh x$$
$$\frac{d}{dx}(\tanh x) = \mathrm{sech}^2 x, \quad \frac{d}{dx}(\coth x) = -\mathrm{csch}^2 x$$

Rules of Differentiation

Use when: The function you're differentiating is a product, quotient, or constant multiple of two simpler functions rather than a single named function above.

$\frac{d}{dx}(cu) = c\frac{du}{dx}$ – constant multiple rule.

$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$ – product rule; use whenever two functions of x are multiplied together.

$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$ – quotient rule; use when one function is divided by another.

Parametric Differentiation

Use when: x and y are both given in terms of a third variable t (common in curve-tracing and motion problems) instead of y being given directly as a function of x.

If $x = x(t)$, $y = y(t)$: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$

Chain Rule

Use when: The function is a "function of a function," e.g. $\sin(x^2)$ or $e^{\cos x}$ – you must differentiate the outer function first, then multiply by the derivative of the inner function.

If $y = f(u)$, $u = g(x)$: $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$

2. Standard Integrals

> **Use when:** You need to reverse a derivative – i.e. find the antiderivative of a standard function. These are the building blocks for every integration/area/volume problem. Always add the constant C.

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}, \qquad \int \frac{1}{x}dx = \log x$$

$$\int e^x\,dx = e^x, \qquad \int a^x\,dx = \frac{a^x}{\log a}$$

$$\int \sin x\,dx = -\cos x, \qquad \int \cos x\,dx = \sin x$$

$$\int \tan x\,dx = \log(\sec x), \qquad \int \cot x\,dx = \log(\sin x)$$

$$\int \sec x\,dx = \log(\sec x + \tan x)$$

$$\int \csc x\,dx = \log(\csc x - \cot x)$$

$$\int \sec^2 x\,dx = \tan x, \qquad \int \csc^2 x\,dx = -\cot x$$

$\displaystyle\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\tan^{-1}\frac{x}{a}$ – use when the denominator is a sum of two squares.

$\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^{-1}\frac{x}{a}$ – use when you see a square root of (constant^2 $-$ x^2).

$\displaystyle\int \frac{f'(x)}{f(x)}dx = \log f(x)$ – use when the numerator is exactly the derivative of the denominator.

$\displaystyle\int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)}$ – same idea, but under a square root.

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx]$$

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx]$$

> **Use when:** The above two: use whenever an exponential is multiplied by a sine or cosine – very common in RL/RC circuit and damped-oscillation problems.

Definite Integral Properties

> **Use when:** Use these as shortcuts to simplify a definite integral *before* you actually integrate – especially useful when the integrand is symmetric or when direct integration looks messy. Always check first whether $f(x)$ is even, odd, or has a useful symmetry about the midpoint of the interval.

$\displaystyle\int_0^a f(x)dx = \int_0^a f(a-x)dx$ – use when replacing $x \to a-x$ makes the integral easier (classic trick for definite integrals with trig functions on $[0,\pi/2]$ or $[0,\pi]$).

$$\int_0^a f(x)dx = -\int_a^0 f(x)dx$$

$\displaystyle\int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & f \text{ even} \\ 0, & f \text{ odd} \end{cases}$ – always check even/odd first on a symmetric interval $[-a,a]$; it can immediately make the integral zero.

Integration by Parts

> **Use when:** The integrand is a *product* of two different types of functions (e.g. polynomial × trig, polynomial × exponential, or a log/inverse-trig function alone) and no substitution simplifies it directly.

$$\int u\,v\,dx = u\int v\,dx - \int\left(\frac{du}{dx}\int v\,dx\right)dx$$

Choose u using ILATE priority (Inverse trig, Log, Algebraic, Trig, Exponential) – whichever type appears first in this list should be taken as u.

Bernoulli's Rule (Repeated Integration by Parts)

> **Use when:** The first function u is a polynomial (so its higher derivatives eventually vanish) and the integral of the second function v is easy to find repeatedly. This avoids doing integration by parts over and over manually.

If 1st function is polynomial and integration of 2nd function is known:

$$\int uv\,dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \cdots$$

where primes denote successive derivatives of u, and v_k is the k-th repeated integral of v. Typical use: $\int x^3e^{2x}dx$ or $\int x^2\sin 3x\,dx$.

3. Vector Calculus

> **Use when:** Any problem involving direction, displacement, velocity/acceleration in 3D, or checking whether two lines/vectors are parallel or perpendicular.

Position vector: $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$, Magnitude: $|\vec{r}| = \sqrt{x^2+y^2+z^2}$

Dot products of unit vectors: $\hat{\imath}\cdot\hat{\imath} = \hat{\jmath}\cdot\hat{\jmath} = \hat{k}\cdot\hat{k} = 1$; $\hat{\imath}\cdot\hat{\jmath} = \hat{\jmath}\cdot\hat{k} = \hat{k}\cdot\hat{\imath} = 0$

Cross products: $\hat{\imath}\times\hat{\imath} = \vec{0}$; $\hat{\imath}\times\hat{\jmath} = \hat{k}$, $\hat{\jmath}\times\hat{k} = \hat{\imath}$, $\hat{k}\times\hat{\imath} = \hat{\jmath}$

Angle between vectors: $\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|}$ – use whenever you're asked for the angle between two vectors/lines/forces.

Unit vector: $\hat{A} = \dfrac{\vec{A}}{|\vec{A}|}$ – use whenever you need a direction only, with magnitude 1.

Velocity: $\vec{V} = \dfrac{d\vec{s}}{dt}$, Acceleration: $\vec{a} = \dfrac{d^2\vec{s}}{dt^2}$ – use in kinematics problems where position is given as a vector function of time.

Dot product: $\vec{A}\cdot\vec{B} = a_1a_2 + b_1b_2 + c_1c_2$ – gives a scalar; use to test perpendicularity ($\vec{A}\cdot\vec{B} = 0$) or find the angle between vectors.

Cross product: $\vec{A}\times\vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ – gives a vector perpendicular to both $\vec{A}, \vec{B}$; use for area of a parallelogram, or to test parallelism ($\vec{A}\times\vec{B} = \vec{0}$).

Scalar triple product: $\vec{A}\cdot(\vec{B}\times\vec{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ – use for the volume of a parallelepiped, or to test if three vectors are coplanar (value $= 0$).

4. Trigonometric Formulae

Use when: Use these to simplify or rewrite trig expressions before differentiating, integrating, or solving trig equations – especially in Fourier series and ODE problems where products of sines/cosines need to become sums (or vice versa).

Identities

$\sin^2\theta + \cos^2\theta = 1, \quad 1 + \tan^2\theta = \sec^2\theta, \quad 1 + \cot^2\theta = \csc^2\theta$ – use to convert between trig functions or simplify expressions under a square root.

Compound Angles

$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$

$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

– use when you must expand sin/cos/tan of a sum or difference of two angles (e.g. $\sin(x + 30°)$), or derive the angle-between-two-lines/curves formula.

Transformation Formulae

Use when: Use "product-to-sum" when integrating a product of sines/cosines (turns a hard product integral into two easy ones); use "sum-to-product" when simplifying $\sin C \pm \sin D$ or $\cos C \pm \cos D$, common in Fourier coefficient derivations.

$$\sin A\cos B = \tfrac{1}{2}[\sin(A+B) + \sin(A-B)]$$
$$\cos A\sin B = \tfrac{1}{2}[\sin(A+B) - \sin(A-B)]$$
$$\cos A\cos B = \tfrac{1}{2}[\cos(A+B) + \cos(A-B)]$$
$$\sin A\sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$$
$$\sin C + \sin D = 2\sin\left(\tfrac{C+D}{2}\right)\cos\left(\tfrac{C-D}{2}\right)$$
$$\sin C - \sin D = 2\cos\left(\tfrac{C+D}{2}\right)\sin\left(\tfrac{C-D}{2}\right)$$
$$\cos C + \cos D = 2\cos\left(\tfrac{C+D}{2}\right)\cos\left(\tfrac{C-D}{2}\right)$$
$$\cos C - \cos D = -2\sin\left(\tfrac{C+D}{2}\right)\sin\left(\tfrac{C-D}{2}\right)$$

Multiple Angle Formulae

Use when: Use when an equation or integral contains $\sin 2A$, $\cos 2A$, $\sin^2 A$, $\cos^2 A$, $\sin^3 A$, or $\cos^3 A$ and you need to reduce the power or rewrite in terms of a single angle – essential before integrating $\sin^2 x$ or $\cos^3 x$ type terms.

$$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$$
$$\sin^2 A = \frac{1-\cos 2A}{2}, \quad \cos^2 A = \frac{1+\cos 2A}{2}$$
$$\sin^3 A = \tfrac{1}{4}[3\sin A - \sin 3A], \quad \cos^3 A = \tfrac{1}{4}[3\cos A + \cos 3A]$$

Hyperbolic & Euler's Formulae

Use when: Use when the problem involves sinh, cosh, complex exponentials $e^{i\theta}$, or you need to convert between trig and exponential form (common in Fourier transform derivations and complex-number-based ODE solutions).

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \cosh^2\theta - \sinh^2\theta = 1$$

$$e^{i\theta} = \cos\theta + i\sin\theta, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

Logarithmic Formulae

Use when: Use to simplify expressions before differentiating (log differentiation) or before expanding a series (e.g. Maclaurin expansion of $\log(1+\sin x)$ needs $\log(AB) = \log A + \log B$ type splitting).

$\log_e(AB) = \log_e A + \log_e B$, $\log_e\left(\frac{A}{B}\right) = \log_e A - \log_e B$

$\log_e x^n = n\log_e x$, $\log_a B = \dfrac{\log_e B}{\log_e a}$

$\log_a a = 1$, $\log_a 1 = 0$, $\log_e 0 = -\infty$

5. Polar Curves & Curvature

Module 1 – appears almost every exam

Use when: Use this entire section whenever a curve is given in polar form $r = f(\theta)$ (instead of $y = f(x)$) and the question asks for: the angle the tangent makes with the radius vector, the angle between two curves, whether two curves cut at right angles, the pedal equation, or the radius of curvature.

Angle between radius vector and tangent (use this first, on a single curve, before anything else in this section):

$$\tan\phi = r\frac{d\theta}{dr} \quad \text{or} \quad \cot\phi = \frac{1}{r}\frac{dr}{d\theta}$$

Angle of intersection of two curves (use after finding ϕ_1, ϕ_2 for each curve at their common point):

$$|\phi_1 - \phi_2| = \tan^{-1}\left\{\left|\frac{\tan\phi_1 - \tan\phi_2}{1+\tan\phi_1\tan\phi_2}\right|\right\}$$

Orthogonal condition – use when asked to *show two curves cut at right angles*: $|\phi_1 - \phi_2| = \dfrac{\pi}{2}$ or $\tan\phi_1 \cdot \tan\phi_2 = -1$

Pedal (p–r) equation – use when the question explicitly says "find the pedal equation": $p = r\sin\phi$

$$\frac{1}{p^2} = \frac{1}{r^2}\left(1+\cot^2\phi\right) = \frac{1}{r^2}\left(1+\frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right)$$

– this is the working form: find $\cot\phi = \frac{1}{r}\frac{dr}{d\theta}$ first, substitute, then eliminate θ using the original curve equation.

Radius of Curvature

Use when: Use Cartesian form when y is given directly as a function of x; parametric form when x, y are both functions of a parameter t (e.g. cycloid); polar form when the curve is given as $r = f(\theta)$; pedal form when the pedal equation $p = p(r)$ is already known or easy to find.

Cartesian form: $\rho = \dfrac{(1+y_1^2)^{3/2}}{y_2}$

Parametric form: $\rho = \dfrac{(\dot{x}^2+\dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$

Polar form: $\rho = \dfrac{(r^2+r_1^2)^{3/2}}{r^2 - r\,r_2 + 2r_1^2}$

Pedal equation form: $\rho = r\dfrac{dr}{dp}$ – fastest method if p is already expressed in terms of r.

6. Indeterminate Forms

Use when: Use when direct substitution into a limit gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$ (or, after algebraic manipulation, one of these forms) – typical in limit-evaluation questions and when checking continuity/differentiability.

L'Hospital's Rule: if $f(a) = g(a) = 0$ or ∞, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Apply repeatedly if the new limit is still $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Standard limits – memorize these; they show up as sub-steps inside bigger limit problems:

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e, \qquad \lim_{n\to 0} (1+n)^{1/n} = e$$

$$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1, \qquad \lim_{x\to a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

7. Series Expansion

Module 2

Use when: Use Taylor's series when you need to expand $f(x)$ around a *nonzero* point $x = a$ (e.g. "expand near $x = 1$"). Use Maclaurin's series (the special case $a = 0$) when the expansion is about the origin – this is by far the more common exam question (e.g. expand $\sqrt{1 + \sin 2x}$, $\log(\sec x)$, $\log(1 + \sin x)$ up to a given power of x).

Taylor's series about $x = a$:

$$y(x) = y(a) + \frac{(x-a)}{1!} y'(a) + \frac{(x-a)^2}{2!} y''(a) + \frac{(x-a)^3}{3!} y'''(a) + \cdots$$

Maclaurin's series about $x = 0$:

$$y(x) = y(0) + \frac{x}{1!} y'(0) + \frac{x^2}{2!} y''(0) + \frac{x^3}{3!} y'''(0) + \frac{x^4}{4!} y''''(0) + \cdots$$

Procedure: find y, y', y'', y''', y'''' at $x = 0$ one by one, then substitute. Watch for functions where differentiating directly gets messy (e.g. $\log(\sec x)$) – sometimes it's faster to expand the inner function first and use known series (like $\log(1+t)$ or $(1+t)^{1/2}$).

8. Composite Functions & Partial Derivatives

Use when: Use when u (or z) is a function of two or more variables which are themselves functions of other variables, and you're asked to prove a relation between partial derivatives (a very common "prove that..." question type in Module 2).

If $z = f(x, y)$, $x = \phi(t)$, $y = \psi(t)$ – use when both x, y depend on a single parameter t:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

If $z = f(x, y)$, $x = \phi(u, v)$, $y = \psi(u, v)$ – use when x, y each depend on two new variables u, v (used to derive Jacobians):

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

If $u = f(r, s, t)$ and r, s, t are functions of x, y, z – use for 3-variable chain rule proofs (e.g. spherical/cylindrical coordinate Jacobians):

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x} \quad \text{(similarly for } y, z\text{)}$$

9. Fourier Series

Use when: Use whenever you must represent a periodic function $f(x)$ (e.g. a square wave, triangular wave, or piecewise-defined periodic function) as an infinite sum of sines and cosines. First identify the interval type and whether $f(x)$ is even, odd, or neither – this decides which shortcut formula below to use.

Even/Odd test (*always check this first* – it can eliminate half the work): $f(-x) = f(x)$ (even); $f(-x) = -f(x)$ (odd)

$$\int_{-l}^{l} f(x)dx = \begin{cases} 2\int_0^l f(x)dx, & f \text{ even} \\ 0, & f \text{ odd} \end{cases}$$

General Fourier series on $(a, a+2l)$ – use this generic form when the interval doesn't match one of the standard cases below:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{l}x\right)$$

where

$$a_0 = \frac{1}{l}\int_a^{a+2l} f(x)dx$$

$$a_n = \frac{1}{l}\int_a^{a+2l} f(x)\cos\left(\frac{n\pi}{l}x\right)dx$$

$$b_n = \frac{1}{l}\int_a^{a+2l} f(x)\sin\left(\frac{n\pi}{l}x\right)dx$$

Key Intervals – pick based on the interval given in the question

- $(0, 2l)$: same formulae, limits 0 to $2l$ – use when the function is defined on $(0, 2l)$ with no symmetry given.
- $(0, 2\pi)$: replace $\frac{n\pi}{l}x$ with nx, limits 0 to 2π – use when period is 2π.
- $(-l, l)$: limits $-l$ to l – use together with the even/odd shortcuts below.
- $(-\pi, \pi)$: limits $-\pi$ to π, use nx – same idea, period 2π, symmetric interval.

Even function on $(-l, l)$ (use to skip computing b_n entirely – it's always 0): $b_n = 0$;

$$a_0 = \frac{2}{l}\int_0^l f(x)dx, \qquad a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\tfrac{n\pi}{l}x\right)dx$$

Odd function on $(-l, l)$ (use to skip computing a_0, a_n – both are 0): $a_0 = 0$, $a_n = 0$;

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\left(\tfrac{n\pi}{l}x\right)dx$$

Half-Range Series on $(0, l)$

Use when: Use when the function is only defined on half the interval $(0, l)$ and you're told to find a *cosine* series (extend as even) or a *sine* series (extend as odd) – the question will specify which one.

Cosine series: $f(x) = \dfrac{a_0}{2} + \sum a_n \cos\left(\dfrac{n\pi}{l}x\right)$

$$a_0 = \frac{2}{l}\int_0^l f\,dx, \qquad a_n = \frac{2}{l}\int_0^l f\cos\left(\tfrac{n\pi}{l}x\right)dx$$

Sine series: $f(x) = \sum b_n \sin\left(\dfrac{n\pi}{l}x\right)$

$$b_n = \frac{2}{l}\int_0^l f\sin\left(\tfrac{n\pi}{l}x\right)dx$$

Practical Harmonic Analysis

Use when: Use when $f(x)$ is given as a *table of numerical values* (not a formula) at equally spaced points – common in engineering data problems. You compute averages from the table instead of doing integration.

$$a_0 = 2[\overline{y}]$$
$$a_n = 2\left[\overline{y\cos\left(\tfrac{n\pi}{l}x\right)}\right], \qquad b_n = 2\left[\overline{y\sin\left(\tfrac{n\pi}{l}x\right)}\right]$$

(bars denote average values from the table); constant term $= \dfrac{a_0}{2}$; first harmonic $= a_1\cos x + b_1 \sin x$; second harmonic $= a_2\cos 2x + b_2\sin 2x$.

10. Fourier Transforms

Use when: Use when the function is defined on the *entire real line* or a semi-infinite interval $(0, \infty)$ rather than being periodic – Fourier transforms generalize Fourier series to non-periodic functions (common in signal-processing style questions).

Infinite FT: $F[f(x)] = F(s) = \displaystyle\int_{-\infty}^{\infty} f(x)e^{isx}dx$; Inverse: $f(x) = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty} F(s)e^{-isx}ds$

Fourier Cosine Transform – use when $f(x)$ is even or only defined for $x \geq 0$ and you want a cosine-based transform: $F_c(s) = \displaystyle\int_0^{\infty} f(x)\cos sx\,dx$; Inverse: $f(x) = \dfrac{2}{\pi}\displaystyle\int_0^{\infty} F_c(s)\cos sx\,ds$

Fourier Sine Transform – use when $f(x)$ is odd or defined for $x \geq 0$ and a sine-based transform is asked for: $F_s(s) = \displaystyle\int_0^{\infty} f(x)\sin sx\,dx$; Inverse: $f(x) = \dfrac{2}{\pi}\displaystyle\int_0^{\infty} F_s(s)\sin sx\,ds$

Properties – use these to avoid recomputing a transform from scratch

Linearity: $F[af + bg] = aF[f] + bF[g]$ – use when $f(x)$ is a sum of two known functions.

Scaling: $F[f(ax)] = \dfrac{1}{a}F\left(\dfrac{s}{a}\right)$ – use when the argument of f is scaled by a constant.

Shifting: $F[f(x-a)] = e^{isa}F(s)$ – use when the function is shifted horizontally.

Modulation: $F[f(x)\cos ax] = \frac{1}{2}[F(s+a) + F(s-a)]$ – use when $f(x)$ is multiplied by a cosine.

11. Z-Transforms

Use when: Use for discrete sequences u_n (instead of continuous functions) – the discrete analogue of the Laplace transform, common in digital-signal / difference-equation problems.

Definition: $Z(u_n) = \displaystyle\sum_{n=0}^{\infty} u_n z^{-n} = U(z)$

u_n	$U(z)$
a^n	$\dfrac{z}{z-a}$
1	$\dfrac{z}{z-1}$
k	$\dfrac{kz}{z-1}$
n	$\dfrac{z}{(z-1)^2}$
n^2	$\dfrac{z^2+z}{(z-1)^3}$
$\cos n\theta$	$\dfrac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$
$\sin n\theta$	$\dfrac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$
$a^n\cos n\theta$	$\dfrac{z(z-a\cos\theta)}{z^2 - 2az\cos\theta + a^2}$
$a^n\sin n\theta$	$\dfrac{az\sin\theta}{z^2 - 2az\cos\theta + a^2}$

Look up the row matching your sequence's pattern – don't derive from the definition unless asked to.

Linearity: $Z(au_n + bv_n - cw_n) = aZ(u_n) + bZ(v_n) - cZ(w_n)$ – use when u_n is a combination of known sequences.

Damping: $Z(a^{-n}u_n) = U(az)$, $\quad Z(a^n u_n) = U(z/a)$ – use when the sequence has an extra a^n or a^{-n} factor multiplied onto a known sequence.

Shifting: $Z(u_{n-k}) = z^{-k}U(z)$, $\;k > 0$ – use when the sequence is delayed/shifted, typical in solving difference equations.

$Z(u_{n+1}) = z[U(z) - u_0]$

$Z(u_{n+2}) = z^2[U(z) - u_0 - u_1 z^{-1}]$ – use these when solving a difference equation involving u_{n+1} or u_{n+2} terms (analogous to Laplace transforms of derivatives for ODEs).

Initial value: $u_0 = \displaystyle\lim_{z\to\infty} U(z)$ – use to find u_0 directly from $U(z)$ without inverting.

Final value: $\displaystyle\lim_{n\to\infty} u_n = \lim_{z\to 1}(z-1)U(z)$ – use to find the long-term/steady-state value of the sequence directly from $U(z)$.

12. Linear Differential Equations (Higher Order)

Use when: Use for ODEs of the form $f(D)y = \phi(x)$ with constant coefficients and order ≥ 2 – solve by finding the Complementary Function (C.F.) from the roots of the auxiliary equation, then a Particular Integral (P.I.) matching the type of $\phi(x)$.

Solution of $f(D)y = \phi(x)$ is $y = \text{C.F.} + \text{P.I.}$

Rules for C.F. – pick the row matching the roots of $f(m) = 0$

Roots of $f(m) = 0$	C.F.
Real, distinct $m_1, \dots, m_k$	$c_1 e^{m_1 x} + \cdots + c_k e^{m_k x}$
Real, repeated (r times)	$(c_1 + c_2 x + \cdots + c_r x^{r-1}) e^{mx}$
Complex $\alpha \pm i\beta$	$e^{\alpha x}[c_1 \cos\beta x + c_2 \sin\beta x]$
Complex, repeated twice	$e^{\alpha x}[(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x]$

Rules for P.I. – pick the row matching the type of $\phi(x)$

For $\phi(x) = e^{ax}$:

$$\frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)} \; (f(a) \neq 0)$$

$$= \frac{xe^{ax}}{f'(a)} \; (f(a) = 0, f'(a) \neq 0) = \frac{x^2 e^{ax}}{f''(a)} \; (\text{and so on})$$

– use the "and so on" rule (differentiate the denominator, multiply numerator by x) whenever the plain substitution gives $\frac{}{0}$.

For $\phi(x) = \sin ax$ or $\cos ax$:

$$\frac{1}{f(D^2)} \sin ax = \frac{\sin ax}{f(-a^2)}$$

$$\frac{1}{f(D^2)} \cos ax = \frac{\cos ax}{f(-a^2)} \quad (f(-a^2) \neq 0)$$

If $f(-a^2) = 0$ (i.e. plain substitution fails – this is the "resonance" case): multiply by x and differentiate denominator: $x \cdot \frac{\sin ax}{f'(-a^2)}$ or $x \cdot \frac{\cos ax}{f'(-a^2)}$

For $\phi(x) = e^{ax}V$ where V is any function of x – use the exponential shift theorem:

$$\frac{1}{f(D)} e^{ax} V = e^{ax} \cdot \frac{1}{f(D + a)} V$$

For $\phi(x) = xV$ – use when x multiplies another function V:

$$\frac{1}{f(D)} xV = \left[x - \frac{f'(D)}{f(D)} \right] \frac{1}{f(D)} V$$

For $\phi(x) = x^m$ (a pure polynomial) – expand $[1 + \phi(D)]^{-1}$ using the binomial series and stop once derivatives of x^m become zero:

$$\frac{1}{f(D)} x^m = [1 + \phi(D)]^{-1} x^m$$

Cauchy's Linear Equation (order n)

Use when: Use when you see the pattern $x^n y^{(n)}, x^{n-1} y^{(n-1)}, \dots$ – i.e. the power of x matches the order of the derivative multiplying it exactly.

$$a_0 x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \cdots + a_n y = \phi(x)$$

Substitute $x = e^z$ (i.e. $z = \log x$); then $xy' = Dy$, $x^2 y'' = D(D-1)y$, $\dots$, where $D = \frac{d}{dz}$. This converts the equation into a constant-coefficient linear ODE in z, solvable using Section 12's C.F./P.I. rules.

Legendre's Linear Equation (order n)

Use when: Same idea as Cauchy's equation, but the variable is a linear expression $(ax + b)$ instead of plain x.

$$a_0 (ax + b)^n y^{(n)} + a_1 (ax + b)^{n-1} y^{(n-1)} + \cdots + a_n y = \phi(x)$$

Substitute $ax + b = e^z$; then $(ax + b)y' = aDy$, $(ax + b)^2 y'' = a^2 D(D-1)y$, $\dots$

13. Curve Fitting, Correlation & Regression

Use when: Use curve fitting when you're given a table of (x, y) data points and must find the best-fit equation of a stated type (straight line, parabola, or power curve) by minimizing squared error – solve the "normal equations" simultaneously for the unknown constants.

Straight line $y = a + bx$ – use when a scatter plot of the data looks roughly linear:

$$\sum y = na + b \sum x, \qquad \sum xy = a \sum x + b \sum x^2$$

Parabola $y = a + bx + cx^2$ – use when the data shows curvature (one bend):

$$\sum y = na + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

Curve $y = a \cdot x^b$ – use when the data follows a power-law trend (test by seeing if $\log y$ vs $\log x$ looks linear): let $X = \log_{10} x$, $Y = \log_{10} y$, $A = \log_{10} a$; solve $Y = A + bX$:

$$\sum Y = nA + b \sum X, \qquad \sum XY = A \sum X + b \sum X^2$$

Mean and Standard Deviation

Use when: Use as the basic building blocks before computing correlation, regression, or any hypothesis test below – almost every statistics question starts here.

$$\bar{x} = \frac{\sum x_i}{n}, \qquad \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x_i^2}{n} - \bar{x}^2$$

For frequency data (values grouped into classes with frequencies f_i): $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$, $\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}$

Correlation and Regression

Use when: Use correlation (r) to measure how strongly two variables move together (value between -1 and 1). Use regression lines when you need to *predict* one variable from the other – note there are two different lines depending on which variable you're predicting.

Correlation coefficient: $r = \frac{\sum XY}{\sqrt{\sum X^2 \sum Y^2}}$, where $X = x - \bar{x}$, $Y = y - \bar{y}$

Regression of y on x (use to predict y given x): $y - \bar{y} = r \frac{\sigma_y}{\sigma_x}(x - \bar{x})$

Regression of x on y (use to predict x given y): $x - \bar{x} = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$

$b_{yx} = r\dfrac{\sigma_y}{\sigma_x}$ – slope of the y-on-x line, $b_{xy} = r\dfrac{\sigma_x}{\sigma_y}$ – slope of the x-on-y line.

Angle between regression lines – use to check how strong the linear relationship is (small angle = strong correlation): $\tan\theta = \dfrac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}\cdot\dfrac{1-r^2}{r}$

Standard error of estimate – use to measure how much actual data scatters around the regression line: $S_x = \sigma_x\sqrt{1-r^2}$, $S_y = \sigma_y\sqrt{1-r^2}$

Rank correlation – use instead of ordinary r when data is given as *ranks* rather than numeric values: $\rho = 1 - \dfrac{6\sum d^2}{n(n^2-1)}$, $d = x - y$

14. Probability Distributions

Use when: Use the basic axioms/rules below to compute probabilities of combined or complementary events before moving to a named distribution.

Axioms: $P(E) \geq 0$, $P(S) = 1$, mutually exclusive $\Rightarrow P(E\cup F) = P(E) + P(F)$

General (use for any two events, exclusive or not): $P(E\cup F) = P(E)+P(F)-P(E\cap F)$; $P(E') = 1-P(E)$

Independence test: $P(E\cap F) = P(E)\cdot P(F)$ – use to check if two events don't affect each other.

Bayes' Theorem – use when you're asked for the probability of a *cause* given an observed *effect* (i.e. "reverse" conditional probability, common in diagnostic/quality-control problems): $P(B_i/A) = \dfrac{P(B_i)P(A/B_i)}{\sum P(B_i)P(A/B_i)}$

Discrete Distributions

Use when: Use for a random variable that takes distinct, countable values (like number of successes in trials, or number of arrivals).

Mean: $E(X) = \sum x_i P(x_i)$; Variance $= E(X^2) - [E(X)]^2$

Binomial – use for a fixed number n of independent trials, each with the same success probability p (e.g. tossing a coin 10 times): $P(x) = {}^nC_x p^x q^{n-x}$; Mean $= np$; Variance $= npq$; SD $= \sqrt{npq}$

Poisson – use for the number of rare/random events in a fixed interval of time or space, when n is large and p is small (e.g. number of calls per hour, defects per batch): $P(x) = \dfrac{e^{-m}m^x}{x!}$; Mean = Variance $= m$

Continuous Distributions

Use when: Use for a random variable that can take any value in an interval (like height, time, or measurement error) instead of distinct countable values.

$f(x) \geq 0$; $\int_{-\infty}^{\infty} f(x)dx = 1$; $E(X) = \int_{-\infty}^{\infty} xf(x)dx$

Normal distribution – use for naturally occurring, symmetric, bell-shaped data (heights, measurement errors, exam scores): $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$; standard variate $z = \dfrac{x-\mu}{\sigma}$ (use z with standard normal tables to find probabilities).

Exponential distribution – use for the waiting time until the next random event (e.g. time between arrivals, time to failure of a component): $f(x) = \alpha e^{-\alpha x}$, $x \geq 0$; Mean $= \dfrac{1}{\alpha}$ = SD

Joint Distributions

Use when: Use when you're tracking two random variables together and need to know how they relate/vary together.

$Cov(X,Y) = E(XY) - E(X)E(Y)$ – measures whether X, Y increase/decrease together.

$\rho(X,Y) = \dfrac{Cov(X,Y)}{\sigma_x\sigma_y}$ – normalized version of covariance (like correlation coefficient).

If independent: $E(XY) = E(X)E(Y)$ – use as a quick independence check (if this fails, they're dependent).

15. Sampling & Hypothesis Testing

Use when: Use this whole section when you must decide, from sample data, whether a claim about a population is likely true (hypothesis testing) – the specific test depends on sample size and what you're comparing.

Large sample if $n \geq 30$ (use z-test / normal approximation); otherwise treat as a small sample (use t-test). Normal: 5% outside $\mu \pm 1.96\sigma$; 1% outside $\mu \pm 2.58\sigma$ – use these to build 95%/99% confidence intervals or z-test critical values.

Standard Error

Use when: Use to measure how much a sample statistic (mean or proportion) is expected to vary from the true population value – needed as the denominator in every large-sample z-test.

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad (\sigma \text{ known})$$

$$SE(p_1 - p_2) = \sqrt{PQ\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}, \qquad P = \frac{n_1p_1 + n_2p_2}{n_1+n_2}$$

t-Test

Use when: Use instead of the z-test whenever the sample size is small ($n < 30$) and the population standard deviation is unknown.

Single sample (use to test if a sample mean differs from a claimed population mean μ): $t = \dfrac{\bar{x}-\mu}{\sigma}\sqrt{n}$ or $t = \dfrac{\bar{x}-\mu}{\sigma_s}\sqrt{n-1}$

Two independent samples (use to compare the means of two separate, unrelated small samples):

$$t = \frac{\bar{x}-\bar{y}}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}, \qquad \sigma_s^2 = \frac{(n_1-1)\sigma_x^2 + (n_2-1)\sigma_y^2}{n_1+n_2-2}$$

Paired samples (use when the same subjects are measured twice – e.g. before/after a treatment): $t = \dfrac{\bar{d}}{\sigma/\sqrt{n}}$, $d_i = x_i - y_i$, $\bar{d} = \dfrac{\sum d_i}{n}$

Chi-Square Test

Use when: Use to compare observed frequencies against expected/theoretical frequencies – e.g. testing goodness of fit to a distribution, or independence between two categorical variables in a contingency table.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Degrees of freedom $= n - c$ ($c = 1$ general, 2 Poisson, 3 Normal) – use the right c depending on which distribution you fitted.

F-Test

Use when: Use to compare the *variances* of two independent samples (not their means) – e.g. to check if two machines/processes have equally consistent output.

$$F = \frac{s_1^2}{s_2^2}\ (s_1^2 > s_2^2), \qquad s_1^2 = \frac{\sum(x-\bar{x})^2}{n_1 - 1},\ s_2^2 = \frac{\sum(y-\bar{y})^2}{n_2 - 1}$$

Always put the larger variance on top so $F \geq 1$, then compare with the tabulated critical value.

16. ANOVA (Analysis of Variance)

Use when: Use instead of repeated t-tests when comparing the means of *three or more* groups at once (one-way) or when data is classified by two factors simultaneously, like treatment and block (two-way).

One-Way Classification

Use when: Use when samples are split into groups by a single factor (e.g. comparing average yield across 3 different fertilizers).

Source	SS	df	Mean Sq.
Between	SSC	$c-1$	$MSC = \frac{SSC}{c-1}$
Within	SSE	$N-c$	$MSE = \frac{SSE}{N-c}$
Total	SST	$N-1$	–

$F = \dfrac{MSC}{MSE}$ (larger over smaller of MSC, MSE) – reject H_0 (equal means) if F exceeds the tabulated critical value.

$$SSC = \sum_i \frac{(\sum X_i)^2}{n_i} - \frac{T^2}{N}$$

$$SST = \sum X^2 - \frac{T^2}{N}, \qquad SSE = SST - SSC$$

Two-Way Classification

Use when: Use when data is classified by two factors at once (e.g. yield by both fertilizer type AND field block) and you need to test both factors' effects separately.

Source	SS	df	Mean Sq.
Columns	SSC	$c-1$	$MSC = \frac{SSC}{c-1}$
Rows	SSR	$r-1$	$MSR = \frac{SSR}{r-1}$
Residual	SSE	$(c-1)(r-1)$	$MSE = \frac{SSE}{(c-1)(r-1)}$

$F_C = \dfrac{MSC}{MSE}$ (tests column factor), $F_R = \dfrac{MSR}{MSE}$ (tests row factor) (reciprocate if the ratio is < 1)

$$CF = \frac{T^2}{N}$$

$$SSC = \sum \frac{(\text{col totals})^2}{r} - CF$$

$$SSR = \sum \frac{(\text{row totals})^2}{c} - CF$$

$$SST = \sum (\text{all values})^2 - CF, \qquad SSE = SST - SSC - SSR$$

Compiled from the VTU III Semester Mathematics Handbook (2022 Scheme), with usage notes added to clarify why and when each formula applies. Statistical tables (Normal, t, Chi-square, F) are in the original handbook and omitted here as they are reference tables, not formulae.